

EVIDENCE FOR NON-THERMAL EXCITATION OF ENERGY LEVELS IN ACTIVE BIOLOGICAL SYSTEMS

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The ratio R of the intensities of anti Stokes and Stokes Raman shift lines of 124 cm^{-1} and of 118 cm^{-1} of active *E. coli* B was measured. For an oscillating system in therm equilibrium $R \approx 0.55$ is expected whereas we find $R \approx 1.0$. This shows that this system is excited strongly above thermal excitation in agreement with a previous theoretical conjecture [2–5].

Recent investigation of Raman effect in bacteria has demonstrated [1] that Raman shift lines below 200 cm^{-1} are found in active (metabolizing) synchronised cells whereas broad bands only are exhibited by both resting cells and nutrient solutions. These findings were considered a first step towards confirmation of the theoretical prediction [2–5] that metabolic energy may be used by biological systems to excite coherent vibrations in the region of 10^{11} – 10^{12} Hz.

Details of the spectra found in [1] depend on both the type of bacteria and the type of nutrient. To show that these Raman lines represent excitations beyond thermal equilibrium requires the ratio R of the intensities of anti Stokes: Stokes lines (in an oscillating system) to lie close to $R \approx 1$ when the thermal equilibrium value $R = \exp(-h\nu/kT)$ is considerably smaller (ν = Raman shift).

We have, therefore, measured R for active cells of *E. coli* B (glucose nutrient). Experimental procedures were as in [1] except that the sensitivity setting for the Rayleigh lines was reduced by a factor 200 compared with that of the shift lines. It should also be emphasized that the incubation temperature must be held above 25°C [6].

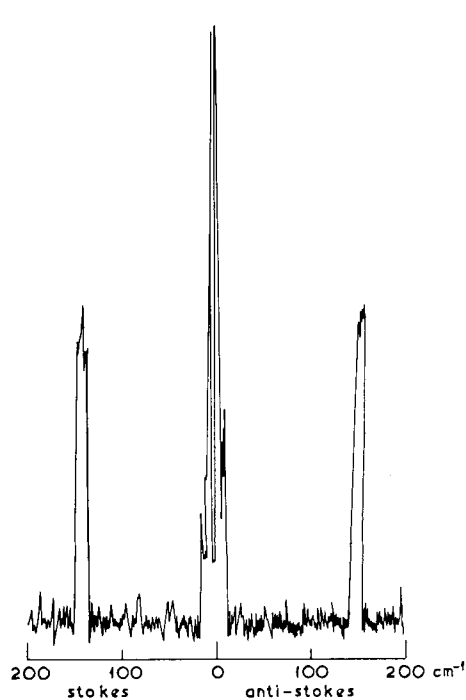
Figs. 1a, b, c give typical results of spectra taken

40, 50, and 60 minutes after begin of incubation. A total of eight series of such spectra was taken so far. We note that the line near 120 cm^{-1} ranges between 118 cm^{-1} and 125 cm^{-1} depending on the age of the cell. At these frequencies the thermal equilibrium value R lies between 0.55 and 0.57. Experimentally, however, we find $R = 1.01 \pm 0.10$ for 118 cm^{-1} and $R = 0.93 \pm 0.13$ for 125 cm^{-1} after averaging over the eight spectra. These values clearly are far above the thermal equilibrium values and thus constitute conclusive confirmation of the theoretical conjecture.

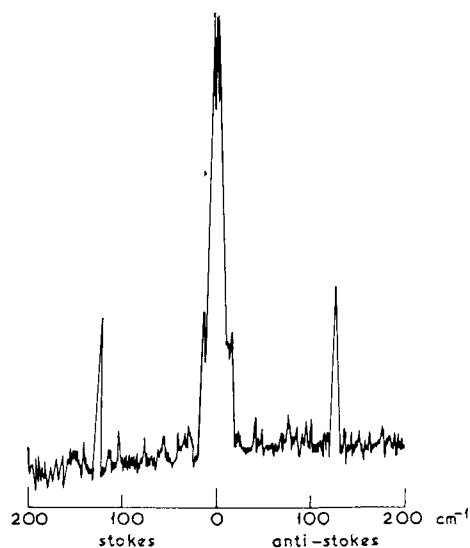
These excitations should be interpreted as arising from certain activities of the cell which change with its age. We note in this context the band near about 40 cm^{-1} which occurs for a short period only.

Finally the lines near 5 cm^{-1} should be noted. They might form a link with biological effects of coherent mm waves (near 2 cm^{-1}) [7].

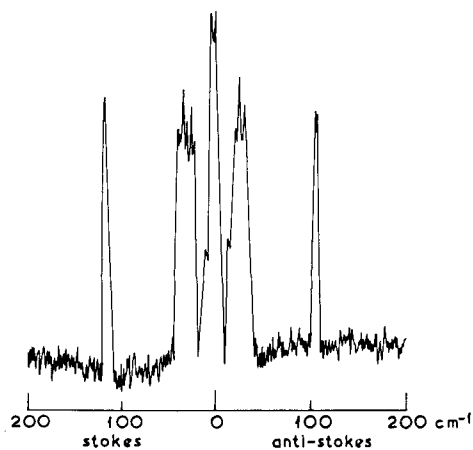
We are grateful to Jane Montgomery for help in the experiments.



(a)



(b)



(c)

Fig. 1a, b, c. Raman spectra of synchronised cells of *E. coli* B taken 40, 50, and 60 minutes after begin of incubation.

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